

AI Ethics

from a computer scientist's perspective

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What is Agency?

Agency

The capacity to make autonomous choices and take intentional actions that affect oneself and the world. Agency requires consciousness, intentionality, and the ability to understand consequences of one's actions.

Core characteristics of agency:

Autonomous decision-making based on goals and values

Intentionality and purposeful action

Understanding and evaluating consequences

Moral responsibility for outcomes

Human Agency vs. AI Systems

Human Agency

Possesses: Consciousness, intentionality, understanding of context and meaning

Creates: Goals, values, and purposes based on lived experience

Bears: Moral responsibility and accountability for decisions and their consequences

Can: Reflect on actions, adapt reasoning, and make ethical judgments

AI Systems (Current)

Lacks: Consciousness, genuine understanding, or subjective experience

Executes: Patterns learned from data without comprehension of meaning or purpose

Cannot: Bear moral responsibility—responsibility remains with human designers and deployers

Appears: Agentic through sophisticated pattern matching, not true agency

Why Agency Matters for AI Ethics

Agency is the organizing principle for applying ethical frameworks to AI systems

1. Responsibility Assignment

Since (current) AI lacks agency, ethical responsibility cannot be delegated to algorithms. Computer scientists and organizations deploying AI must take ownership of all outcomes and impacts.

2. Design as Ethical Act

Every technical decision, from data collection to model architecture to deployment context, embeds ethical choices. These are human decisions with moral weight.

3. Focus on Human Agency

Ethics in AI is fundamentally about how human agency designs, deploys, and governs systems that affect other humans. The question is (currently) not "is the AI ethical?" but "are we being ethical in how we create and use AI?"

The Agency Framework

Three Core Questions for AI Ethics

1

Who has agency?

- Humans (designers, companies, users, regulators)
- AI systems (appear to have agency, but do they really?)
- Both? Neither? (responsibility gaps)

2

Who are the moral subjects versus objects?

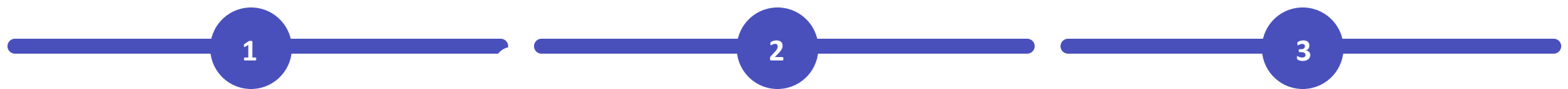
- Subjects (agents with moral responsibility)
- Objects (those affected)
- Confused category (AI treated as subject but is it?)

3

Who should have agency?

- Where should humans maintain control?
- Where can we safely delegate to AI?
- How do we preserve human agency?

Ethics for...?



1

Designing AI systems

- agency with engineers and other stakeholders

2

AI itself – how it should behave

- initially, as AI designers designed it
- agency with AI?

3

Using AI

- agency with you (+ possibly with AI)

Three Ethical Frameworks

When you have agency, how do you reason about ethical decisions? Here are three major philosophical frameworks that provide different lenses for ethical reasoning:

Consequentialism

Core principle: Actions are judged by their outcomes—maximize good, minimize harm

Deontology

Core principle: Rules, duties, and rights matter regardless of outcomes—some things are simply wrong

Virtue Ethics

Core principle: Character and intentions matter—what kind of person are you being?

Virtue Ethics

Core Philosophy

Principle: Character and intentions matter. Focus on virtues, not rules or outcomes

- What kind of person should I be?
- Cultivate virtues: honesty, courage, wisdom, justice
- Character developed through practice

Key thinker: Aristotle

For AI Designers

What kind of culture and character does your design work cultivate?
What virtues guide your decisions?

Example: Building addictive features cultivates greed; accessibility cultivates justice

Strengths: Emphasizes moral development and culture

Limitations: Less clear guidance for specific decisions

Deontology (Kant)

Core Philosophy

Principle: Rules and duties matter regardless of outcomes

- Some actions are wrong in themselves
- Respect for human dignity and autonomy
- Treat people as ends, never merely as means

Key thinker: Immanuel Kant

Categorical Imperative: Act only according to maxims you'd want to be universal law

For AI Designers

Some practices are wrong even if beneficial: privacy violations, manipulation, treating users as mere means

Example: Dark patterns that manipulate users are wrong regardless of business outcomes

Strengths: Protects fundamental rights, clear boundaries, human dignity central

Limitations: Rules can conflict; rigid; doesn't help prioritize when rights clash

Consequentialism / Utilitarianism

Core Philosophy

Principle: Actions are judged entirely by their outcomes

- Maximize aggregate good
- Minimize aggregate harm
- "Greatest good for greatest number"

Key thinkers: Jeremy Bentham, John Stuart Mill

Central question: What produces the best overall consequences?

For AI Designers

Weigh total welfare impact of deploying systems: Who benefits, who's harmed, by how much?

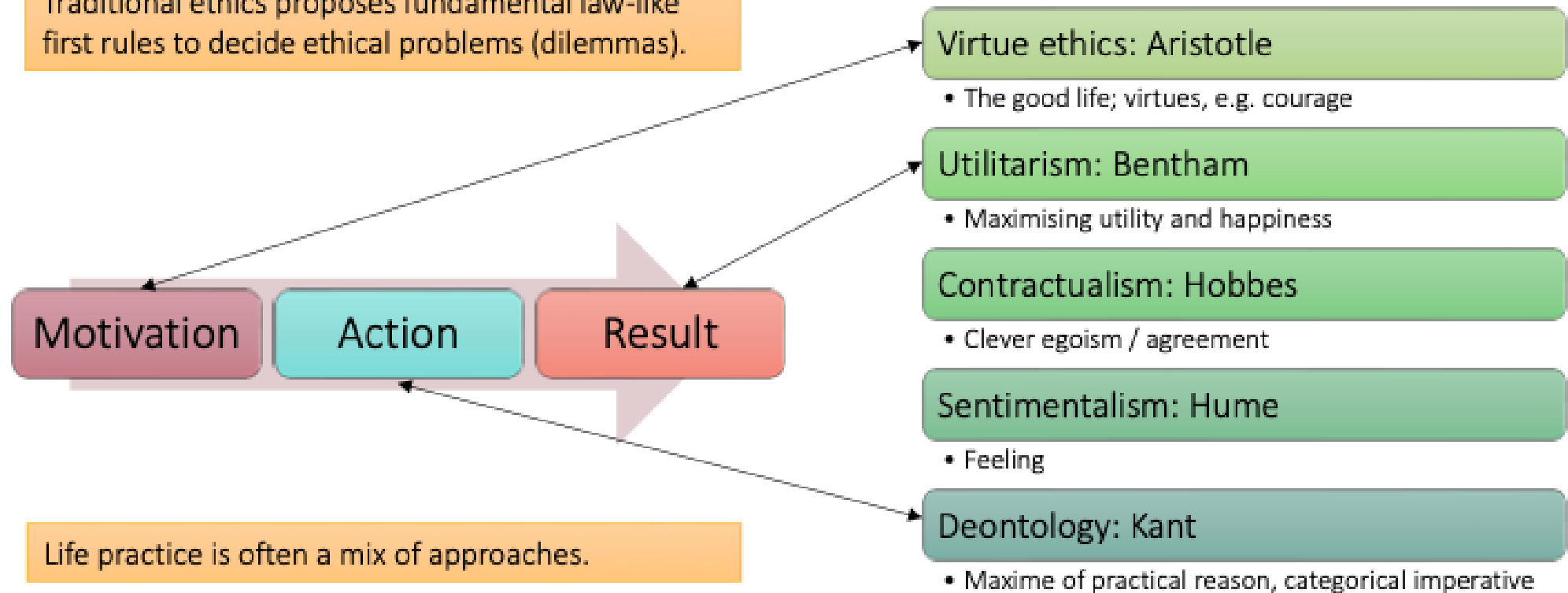
Example: Deploy facial recognition if aggregate benefits outweigh aggregate harms?

Strengths: Practical, measurable, policy-relevant

Limitations: Hard to predict consequences; can justify harming minorities for majority benefit; whose welfare counts?

Types of ethics

Traditional ethics proposes fundamental law-like first rules to decide ethical problems (dilemmas).



The Trolley Problem

Utilitarianism

→ PULL THE LEVER

- Five deaths versus one death
- Minimise total harm
- Clear utilitarian calculation
- Save more lives

The answer appears obvious: aggregate welfare demands pulling the lever.

The Trolley Problem

Deontology

→ DON'T PULL

- Pulling actively kills (not merely letting die)
- Uses person as means to save others
- Violates duty not to kill
- Action itself wrong, regardless of outcome

Even if the outcome is "better," the act of killing violates fundamental duty.

The Trolley Problem

Virtue Ethics

→ IT DEPENDS

- What would a virtuous person do?
- Is pulling courageous or cowardly?
- Does it show care or callousness?
- Requires practical wisdom (phronesis)

No formula provides the answer, only developed character and practical wisdom.

Principlism

Rather than choosing one ethical framework, **principlism** identifies widely-shared ethical principles that can guide practical decision-making across different philosophical traditions.

Core Idea

Focus on mid-level principles that bridge philosophical theory and practical action

Origins in Biomedical Ethics

Developed by Beauchamp & Childress (1979) for medical ethics, now widely applied to AI and technology ethics

Practical Advantage

Actionable guidance without requiring commitment to a single philosophical framework, useful for multidisciplinary teams

Key Principles in AI Ethics

Fairness

Treat individuals and groups equitably; avoid discrimination and unjust bias

Transparency

Make AI systems understandable and explainable to appropriate stakeholders

Accountability

Clear assignment of responsibility for AI system outcomes and impacts

Privacy

Protect personal data and respect individuals' informational self-determination

Safety & Security

Ensure systems are robust, reliable, and secure against misuse or failure

Human Autonomy

Preserve human agency, choice, and meaningful control over AI systems

These principles often conflict in practice: fairness vs. accuracy, privacy vs. transparency, autonomy vs. safety. Trade-offs must be taken. Principles found across EU AI Act, UNESCO recommendations, IEEE standards, and major AI ethics guidelines.

When Principles Conflict

Principles rarely align...

Common Trade-offs in AI Systems

Fairness ↔ Accuracy

Improving fairness across groups often reduces overall accuracy. Which matters more? For whom?

Privacy ↔ Transparency

Explaining how an AI works may reveal sensitive data. How much transparency can you provide without compromising privacy?

Autonomy ↔ Safety

Preventing harmful choices restricts user autonomy. Should the system allow errors to preserve freedom?

Efficiency ↔ Accountability

Full human oversight slows decisions. How much automation can you introduce while maintaining responsibility?